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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 07/02/2015 |
| **PI Name:** | Rubin, David M.S. | | |
| **Project Title:** | Electronic Medical Records for Exploration Medical System | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Operational and clinical research | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **ExMC**:Exploration Medical Capabilities | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77058 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 07/01/2011 | **End Date:** | 03/27/2015 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 3/27/2015 (from 9/30/2018) per PI (Ed., 6/30/15) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

The Electronic Medical Record (EMR) is an electronic system that provides a repository for medical data, images, videos, and reports and makes the information accessible to the crew in flight via a graphical user interface to assist in the diagnosis and treatment of acute and chronic medical conditions.
The system will be used to access medical data for all medical encounters, especially during periods of exploration missions when contact with ground resources (e.g., flight surgeon) will either be minimal or absent. During these periods of autonomous medical care, the crew will need such a system for urgent and non-urgent medical events.

Development of the EMR addresses the following risk as stated within the Critical Path Roadmap (CPR; also known as the NASA Bioastronautics Roadmap (NASA/SP-2004-6113)):

Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities: Given that

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| **Task Description:** | medical capabilities will be limited during human exploration missions, there is a possibility that in-flight medical events will lead to unacceptable health and mission outcomes.<br><br>The objectives of the EMR are to:<br><br>a. Reduce or eliminate the time required of an on-orbit crew (i.e., clinician and non-clinician) to store, access, and process medical data, images, and reports.<br><br>b. Reduce the time required of the ground crew to access medical data, images, or reports.<br><br>c. Ensure Crew Medical Office (CMO) access to all relevant crewmember medical data, images, or reports for improved quality of care.<br><br>d. Ensure crew access to personal medical data, images, or reports during periods of restricted ground communication.<br><br>e. Ensure crew ability to capture onboard medical data regardless of the state of the communication link to the ground.<br><br>f. Ensure privacy protection of onboard crewmember medical data.<br><br>The ground demonstration of the EMR, as part of the Exploration Medical System Demonstration (EMSD), occurred in February, 2015. |
| **Rationale for HRP Directed Research:** | This study is a combination of a solicited task (RFP released under Wyle for procurement of COTS EMR system) and a directed task involving customization of COTS software for the specific needs of ExMC. The study team is uniquely positioned to perform this function and to ensure that the EMR component of the overall Exploration Medical System Demonstration (EMSD) is integrated with the other components of the system. |
| **Research Impact/Earth Benefits:** | Development of the OpenEMR application programming interfaces (APIs), required for integration into the Exploration Medical System, can be made available to the OpenEMR open source community for potential inclusion into future releases of the EMR software. These APIs enhance the ability to import and export data into/out of the EMR database. In addition, the Exploration Medical System can serve as software infrastructure for future development and integration of medical devices and software. |
| **Task Progress:** | All required development of the Electronic Medical Record for integration into the greater Exploration Medical System Demonstration project has been completed. This includes the development of APIs, data forms, and associated intermediate software "adapters" and databases. The performance of the Electronic Medical Record was tested during the EMSD Ground Demonstration in February, 2015. |
| **Bibliography Type:** | Description: (Last Updated: 04/16/2014) |